

MASTER OF SCIENCE

COURSE PLAN

DISCRETE MATHEMATICAL STRUCTURES

Prepared by

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MASTER OF SCIENCE

UNIT-WISE COURSE PLAN

DISCRETE MATHEMATICAL STRUCTURES

Total Hours: 60 (15 weeks $\times$ 4 hours)

Text Book: Kenneth H. Rosen, *Discrete Mathematics & Its Applications with Combinatorics and Graph Theory*, Tata McGraw-Hill

UNIT I: The Foundations – Logic, Proofs and Basic Structures — 11 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Propositional Logic – propositions, connectives, conditional and biconditional statements, truth tables	1	Lecture / Board	Translate English sentences into logical expressions; build truth tables
2	Propositional Equivalences – tautology, contradiction, logical equivalence laws, normal forms (DNF/CNF)	1	Lecture / Worked Examples	Prove equivalences using laws and truth tables
3	Predicates and Quantifiers – universal and existential quantifiers, negation of quantified statements	1	Lecture / Worked Examples	Express statements using predicates and quantifiers
4	Nested Quantifiers	1	Problem Solving	Translate and negate nested quantified statements
5	Rules of Inference – valid arguments, inference rules for propositions and quantified statements	2	Lecture / Guided Problem Solving	Check validity of arguments step by step
6	Introduction to Proofs; Proof Methods and Strategy – direct, contraposition, contradiction, cases, existence and uniqueness proofs	2	Worked Examples / Think-Pair-Share	Write proofs of simple number-theoretic statements
7	Sets and Set Operations – Venn diagrams, set identities, power set, Cartesian product	1	Lecture / Board	Prove set identities; solve Venn-diagram problems
8	Functions – one-to-one, onto, bijection, inverse, composition, floor and ceiling functions	1	Lecture / Worked Examples	Classify functions; find inverses and compositions
9	Sequences and Summations – arithmetic and geometric progressions, summation formulae	1	Problem Solving / Quiz	Evaluate summations; short quiz on Unit I

UNIT II: Algorithms, Integers and Matrices; Induction and Recursion — 10 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Algorithms; Growth of Functions – Big-O, Big-Omega and Big-Theta notations	1	Lecture / Board	Find Big-O estimates of given functions
2	Complexity of Algorithms – time complexity of searching and sorting algorithms	1	Lecture / Worked Examples	Analyse linear search, binary search and bubble sort
3	The Integers and Division; Primes and Greatest Common Divisors	1	Lecture / Worked Examples	Apply division algorithm, prime factorisation and GCD/LCM
4	Integers and Algorithms – base-b expansions, Euclidean algorithm, modular exponentiation	1	Demonstration / Problem Solving	Trace the Euclidean algorithm and fast modular exponentiation
5	Applications of Number Theory – linear congruences, Chinese Remainder Theorem, RSA cryptosystem	1	Case Study / Worked Examples	Encrypt and decrypt a small message with RSA
6	Matrices – matrix arithmetic, transpose, zero–one matrices and Boolean product	1	Lecture / Problem Solving	Compute Boolean products of zero–one matrices
7	Mathematical Induction	1	Worked Examples / Guided Practice	Prove summation formulae and inequalities by induction
8	Strong Induction and Well-Ordering	1	Lecture / Problem Solving	Prove results needing strong induction (e.g. postage problems)
9	Recursive Definitions and Structural Induction	1	Lecture / Worked Examples	Define sets and functions recursively; prove properties by structural induction
10	Recursive Algorithms and Program Correctness – partial correctness, loop invariants	1	Problem Solving / Assignment	Write recursive algorithms; verify a loop using invariants

UNIT III: Counting and Advanced Counting Techniques — 11 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	The Basics of Counting – sum, product, subtraction and division rules, tree diagrams	1	Lecture / Board	Solve counting problems on strings, passwords and licence plates
2	The Pigeonhole Principle – basic and generalised forms	1	Worked Examples	Apply the pigeonhole principle to existence problems
3	Permutations and Combinations	1	Lecture / Problem Solving	Solve arrangement and selection problems
4	Binomial Coefficients – Binomial Theorem, Pascal's identity and triangle, Vandermonde's identity	1	Lecture / Worked Examples	Expand binomials; prove combinatorial identities
5	Generalized Permutations and Combinations; Generating Permutations and Combinations	1	Problem Solving / Demonstration	Count with repetition; generate next permutation in lexicographic order
6	Recurrence Relations – modelling with recurrences (Fibonacci, Towers of Hanoi, bit strings)	1	Lecture / Worked Examples	Set up recurrence relations for counting problems
7	Solving Linear Recurrence Relations – homogeneous and non-homogeneous with constant coefficients	2	Worked Examples / Guided Problem Solving	Solve recurrences using characteristic roots and particular solutions
8	Divide-and-Conquer Algorithms and Recurrence Relations – Master Theorem	1	Lecture / Problem Solving	Derive recurrences for binary search and merge sort
9	Generating Functions – solving counting problems and recurrences	1	Lecture / Worked Examples	Find coefficients of generating functions for counting problems
10	Inclusion–Exclusion and Applications of Inclusion–Exclusion – onto functions, derangements, sieve of Eratosthenes	1	Problem Solving / Quiz	Count derangements and onto functions; quiz on Unit III

UNIT IV: Relations — 6 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Relations and their Properties – reflexive, symmetric, antisymmetric, transitive; combining relations	1	Lecture / Board	Test given relations for each property
2	n-ary Relations and their Applications – relational databases, selection, projection and join	1	Case Study / Demonstration	Perform select, project and join on a sample student database
3	Representing Relations – zero–one matrices and directed graphs	1	Lecture / Worked Examples	Convert relations between matrix and digraph forms
4	Closures of Relations – reflexive, symmetric and transitive closures; Warshall's algorithm	1	Worked Examples / Guided Practice	Compute transitive closure using Warshall's algorithm
5	Equivalence Relations – equivalence classes and partitions	1	Lecture / Problem Solving	Find equivalence classes, e.g. congruence modulo n
6	Partial Orderings – posets, Hasse diagrams, maximal/minimal elements, lattices, topological sorting	1	Problem Solving / Quiz	Draw Hasse diagrams; topologically sort a task list; quiz on Unit IV

UNIT V: Graphs and Trees — 11 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Graphs and Graph Models; Graph Terminology and Special Types of Graphs – degree, handshaking theorem, complete, cycle, wheel, bipartite graphs	1	Lecture / Board	Identify graph types; verify the handshaking theorem
2	Representing Graphs and Graph Isomorphism – adjacency list, adjacency and incidence matrices	1	Worked Examples	Represent graphs; test pairs of graphs for isomorphism
3	Connectivity – paths, connected components, cut vertices and cut edges	1	Lecture / Problem Solving	Find components and cut vertices; count paths using adjacency matrix
4	Euler and Hamilton Paths and Circuits	1	Lecture / Worked Examples	Check Euler and Hamilton conditions on given graphs
5	Shortest-Path Problems – Dijkstra's algorithm, travelling salesperson problem	1	Demonstration / Guided Practice	Trace Dijkstra's algorithm on a weighted graph
6	Planar Graphs – Euler's formula, Kuratowski's theorem	1	Lecture / Problem Solving	Apply Euler's formula; show K ₅ and K _{3,3} are non-planar
7	Graph Colouring – chromatic number, applications to scheduling	1	Problem Solving / Think-Pair-Share	Colour graphs; model an exam-timetable problem
8	Introduction to Trees; Applications of Trees – binary search trees, decision trees, prefix codes, Huffman coding	1	Lecture / Worked Examples	Build a BST; construct Huffman codes
9	Tree Traversal – preorder, inorder, postorder; infix, prefix and postfix notation	1	Worked Examples	Traverse trees; evaluate prefix and postfix expressions
10	Spanning Trees – construction using DFS and BFS, backtracking	1	Demonstration / Guided Practice	Find spanning trees by DFS and BFS
11	Minimum Spanning Trees – Prim's and Kruskal's algorithms	1	Problem Solving / Quiz	Find MSTs by both algorithms and compare; quiz on Unit V

UNIT VI: Boolean Algebra and Modelling Computation — 8 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Boolean Functions – Boolean expressions, identities of Boolean algebra, duality	1	Lecture / Board	Prove Boolean identities; find duals
2	Representing Boolean Functions (sum-of-products, functional completeness); Logic Gates	1	Lecture / Worked Examples	Write SOP expansions; draw gate circuits for given functions
3	Minimization of Circuits – Karnaugh maps and the Quine–McCluskey method	2	Worked Examples / Guided Problem Solving	Minimise functions of 3 and 4 variables
4	Languages and Grammars – phrase-structure grammars, types of grammars, derivation trees	1	Lecture / Worked Examples	Derive strings and classify grammars
5	Finite-State Machines with Output – Mealy and Moore machines	1	Demonstration / Problem Solving	Design a vending-machine or unit-delay machine
6	Finite-State Machines with No Output; Language Recognition – DFA, NFA, regular sets, Kleene's theorem	1	Lecture / Worked Examples	Construct DFAs for given languages; convert NFA to DFA
7	Turing Machines – definition, computing with Turing machines	1	Lecture / Quiz	Trace a Turing machine on an input tape; quiz on Unit VI

UNIT VII: Revision and Assessment — 3 Hours

S.No.	Topic	Hrs	Mode of Teaching	Learning Activity
1	Revision of Units I–III – logic, proofs, number theory, induction and counting	1	Tutorial / Problem Solving	Solve previous university question papers
2	Revision of Units IV–VI – relations, graphs, trees, Boolean algebra and FSMs	1	Tutorial / Problem Solving	Solve previous university question papers
3	Assignment review, slip test and doubt clearance	1	Slip Test / Discussion	Written test and feedback on assignments

Overall Hour Distribution

Unit	Unit Title	Hours	Percentage
I	The Foundations – Logic, Proofs and Basic Structures	11	18.33%
II	Algorithms, Integers and Matrices; Induction and Recursion	10	16.67%
III	Counting and Advanced Counting Techniques	11	18.33%
IV	Relations	6	10.00%
V	Graphs and Trees	11	18.33%
VI	Boolean Algebra and Modelling Computation	8	13.33%
VII	Revision and Assessment	3	5.00%
	GRAND TOTAL	60	100%

General Modes of Teaching

- Each session is a 1-hour lecture: concepts are introduced on the board or through PPT, followed by worked examples from the text book.
- Every unit includes guided problem-solving hours where students solve exercises in class with faculty support.
- Active-learning techniques such as Think-Pair-Share and short quizzes check understanding at the end of each unit.
- Algorithmic topics (Euclidean algorithm, Warshall's, Dijkstra's, Prim's, Kruskal's, K-maps, FSMs) are demonstrated by step-by-step tracing on sample inputs.
- Case studies such as RSA cryptography, relational databases and exam timetabling by graph colouring connect theory with real-world applications.
- Continuous assessment through unit-wise assignments, quizzes, slip tests and mid-semester examinations.
- Additional problem sheets are given to fast learners, and remedial tutorials are held for slow learners.
- The final three hours are for revision, solving previous question papers and the slip test.